

ApproxDA-TransUNet: Exploring Context-Dependent Attention Approximation in Medical Image Segmentation

Author Name

Department of Computer Science

University Name

City, Country

author@university.edu

Abstract—We introduce Global Context Requirement (GCR), a task-level property capturing the degree to which accurate segmentation depends on long-range spatial dependencies, and propose that approximation safety—the degree to which attention approximation can be introduced without degrading accuracy—is inversely governed by GCR. To validate this hypothesis, we design ApproxDA-TransUNet, a practical efficient dual-attention architecture built around three orthogonal approximation operators—windowed PAM (spatial scope M), low-rank projection (representation fidelity r), and grouped CAM (channel scope G)—reducing per-GPU training memory from 11.5 GB to 6.4 GB and enabling multi-GPU distributed training. Experiments on three benchmarks confirm the GCR hypothesis: the same configuration degrades multi-organ CT segmentation (78.64% DSC on Synapse, -1.16% vs. DA-TransUNet, high GCR) while improving polyp segmentation (89.24% DSC on Kvasir-SEG, $+0.77\%$, low GCR). A secondary finding reveals gradient symmetry collapse: at initialization $g \approx 0.5$, symmetric branch gradients drive convergent representations and near-zero gate updates, pinning $g \approx 0.5$ regardless of window size, confirmed independently at both $M=7$ and $M=14$. These results shift the research question from *how to approximate* to *when to approximate*, with GCR providing a practical task-level criterion.

Index Terms—medical image segmentation, dual attention, efficient transformers, attention approximation, gate collapse, global context

I. INTRODUCTION

Accurate medical image segmentation is critical for clinical workflows in radiology, endoscopy, and dermatology, where precise delineation of anatomical structures drives diagnosis. CNN-based encoder-decoder architectures—including U-Net [1], U-Net++ [2], Attention U-Net [3], ResUNet [4], and MultiResUNet [5]—established the foundational segmentation paradigm using skip connections for spatial detail recovery, yet their limited receptive fields constrain reasoning over long-range anatomical relationships. Transformer-based architectures have since transformed the field by capturing these long-range dependencies through dual attention—jointly modeling positional correlations via Position Attention Modules (PAM) and channel semantic correlations via Channel Attention Modules (CAM) [6]. Representative methods such as DA-TransUNet [7] integrate these into hybrid CNN-Transformer

backbones, demonstrating state-of-the-art segmentation performance across multiple benchmarks. However, both PAM and CAM carry quadratic computational complexity ($\mathcal{O}(N^2C)$ and $\mathcal{O}(C^2N)$, respectively), making high-resolution deployment increasingly expensive.

To address this challenge, a large body of recent research has focused on efficient attention approximations, including window-based attention [8], low-rank factorization [9], sparse attention, linear attention [10], and grouped channel interaction [11]. These approximation strategies have substantially reduced the theoretical computational complexity of attention while maintaining competitive performance in many computer vision tasks. Consequently, efficient attention has become a standard design paradigm for modern vision transformers [12]–[14]. Yet a fundamental assumption underlies all of these approaches: that approximation is universally applicable, with at most minor accuracy trade-offs.

Despite these advances, an important scientific question remains largely unanswered:

When is attention approximation safe for medical image segmentation?

We hypothesize that the answer is governed by a task-level property we call **Global Context Requirement (GCR)**—the degree to which accurate segmentation depends on long-range spatial dependencies. Tasks with high GCR (multi-organ CT) require global anatomical reasoning; tasks with low GCR (polyp, skin lesion) are dominated by local appearance. We predict an inverse relationship: **Approximation Safety** $\approx f(\text{GCR})$, $f' < 0$. To test this hypothesis empirically, we design **ApproxDA-TransUNet**, an efficient dual-attention architecture with three orthogonal approximation operators (windowed PAM, low-rank projection, grouped CAM), each independently controllable, so that any axis can be varied in isolation.

During this investigation, we obtain two unexpected observations.

First, attention approximation exhibits clear task-dependent behavior. Although ApproxDA achieves competitive performance on local-structure-dominant datasets such as Kvasir (and ISIC, if confirmed), the same approximation consistently

degrades performance on the multi-organ Synapse benchmark, providing empirical evidence that high-GCR tasks cannot tolerate aggressive spatial approximation.

Second, the learnable routing mechanism originally introduced to adaptively fuse spatial and channel attention consistently collapses to an almost constant routing coefficient across different datasets and architectural configurations. Further analysis reveals that this phenomenon originates from gradient symmetry between the two attention branches, causing the routing module to degenerate into a nearly fixed fusion strategy. Rather than functioning as sample-adaptive routing, the gate behaves as a task-level fixed approximation whose effectiveness depends primarily on the contextual characteristics of the segmentation task.

Together, these results provide evidence consistent with the GCR hypothesis: the same ApproxDA configuration achieves +0.80% DSC on low-GCR Kvasir-SEG and −1.16% DSC on high-GCR Synapse—opposite effects consistent with Approximation Safety $\approx f(\text{GCR})$.

This paper is organized around three explicit hypotheses:

- **H1:** Attention approximation effectiveness depends on task characteristics and is not universally safe.
- **H2:** Global Context Requirement (GCR)—a latent task property describing the degree to which segmentation relies on long-range spatial dependencies—is an important explanatory factor governing approximation safety.
- **H3:** Symmetric dual-attention routing naturally collapses to a fixed equilibrium during optimization, independent of window size or rank.

The main contributions of this paper are summarized as follows:

- We introduce **Global Context Requirement (GCR)** and **Approximation Safety** as complementary task-level concepts, and propose that Approximation Safety $\approx f(\text{GCR})$ with $f' < 0$. Our experiments provide evidence consistent with this: the same configuration incurs −1.16% DSC on high-GCR Synapse (multi-organ CT, global anatomical reasoning required) while achieving +0.80% DSC on low-GCR Kvasir-SEG (polyp segmentation, local appearance dominant)—opposite effects consistent with GCR as an important explanatory factor.
- We design **ApproxDA-TransUNet**, a practical efficient dual-attention architecture that replaces the quadratic DA module with three orthogonal approximation operators (windowed PAM, low-rank projection, grouped CAM), each independently controllable. The architecture simultaneously achieves 44% lower per-GPU VRAM (6.4 GB vs. 11.5 GB), resolves a DataParallel failure in DA-TransUNet, and enables systematic investigation of how each approximation axis affects performance under different GCR regimes.
- We derive the gradient dynamics of symmetric dual-branch gating and empirically verify **gradient symmetry collapse**: at initialization $g \approx 0.5$, PAM and CAM receive equal gradients and converge to similar representations,

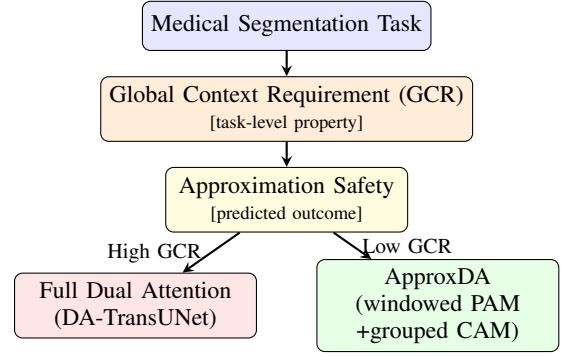


Fig. 1. The GCR hypothesis studied in this work. A segmentation task’s Global Context Requirement (GCR) appears to govern approximation safety and guide attention strategy selection. ApproxDA-TransUNet serves as the controllable experimental instrument for testing this hypothesis; gate collapse is a secondary mechanistic finding.

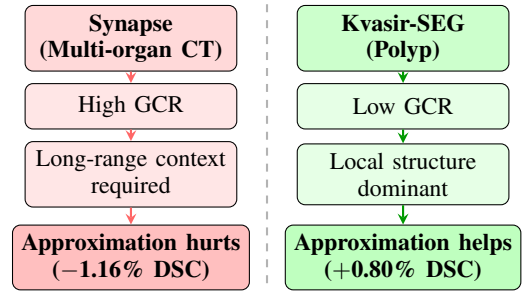


Fig. 2. Context-dependent approximation safety governed by Global Context Requirement (GCR). The same ApproxDA configuration yields opposite effects depending on how much the task relies on long-range spatial context. This is the central empirical finding of this paper.

driving the gate gradient to zero and fixing $g \approx 0.5$ regardless of window size—confirmed independently at both $M=7$ and $M=14$.

The remainder of this paper is organized as follows: Section II reviews related work; Section III presents the ApproxDA framework; Section IV reports experimental results; Section V provides discussion and analysis; Section VI concludes.

II. RELATED WORK

A. CNN-based Medical Image Segmentation

Convolutional encoder-decoder architectures have established the foundational design paradigm for medical image segmentation. U-Net [1] introduced a symmetric encoder-decoder with skip connections that combine deep semantic features with shallow spatial detail, enabling precise boundary localization. U-Net++ [2] densified the skip pathways through nested sub-networks to reduce the semantic gap between encoder and decoder. Attention U-Net [3] integrated soft attention gates on skip connections to selectively suppress irrelevant activations at each scale. ResUNet [4] incorporated residual connections for stable training at greater depth, while MultiResUNet [5] replaced standard convolutions with multi-resolution blocks for richer local feature representations. These five architectures form the primary CNN baselines evaluated in

this work. Despite their strong performance on local-feature-dominant tasks, their limited receptive fields hinder long-range anatomical reasoning, motivating the Transformer-based methods discussed next.

B. Transformer-based Medical Image Segmentation

Transformer architectures have significantly advanced medical image segmentation by enabling global dependency modeling beyond the locality of convolutional neural networks. Representative methods such as TransUNet [12], Swin-Unet [13], UNETR [15], UCTransNet [16], and DAEformer [14] combine convolutional feature extraction with transformer-based contextual modeling to achieve state-of-the-art segmentation performance across a wide range of medical imaging tasks.

C. Dual Attention and DA-TransUNet

Position Attention Modules (PAM) and Channel Attention Modules (CAM) were introduced by DANet [6] for scene segmentation. PAM computes a non-local affinity matrix over all $N=H \times W$ spatial positions at $\mathcal{O}(N^2C)$ complexity, enabling each location to aggregate context from every other position. CAM models dependencies among all C channels through a channel-wise affinity matrix at $\mathcal{O}(C^2N)$ complexity, capturing global semantic channel correlations independent of spatial layout. Outputs of both branches are summed via learned coefficients to produce the final dual-attended representation.

DA-TransUNet [7] adapted this design for medical image segmentation by inserting dual-attention blocks at multiple skip connections and the bottleneck of a hybrid R50-ViT-B/16 encoder-decoder, with a learnable channel-wise gate fusing PAM and CAM at each block. DA-TransUNet achieves state-of-the-art accuracy on multiple benchmarks, but inherits the quadratic complexity of PAM ($\mathcal{O}(N^2C)$) and CAM ($\mathcal{O}(C^2N)$), and its public implementation contains a DataParallel incompatibility that prevents multi-GPU training. This work directly targets the dual-attention module of DA-TransUNet.

D. Adaptive Attention Routing

Learnable gating for feature branch fusion appears widely in transformer-based segmentation [11], [14], with the implicit assumption that such gates converge to meaningful, sample-adaptive strategies after end-to-end optimization. Existing work generally does not investigate whether the routing module remains effective or degenerates. This work explicitly analyzes the optimization dynamics of the gate within DA-TransUNet [7] and shows that gradient symmetry between the two attention branches causes the gate to collapse to a near-constant coefficient, degenerating from adaptive routing into a fixed blend.

E. Research Gap

Existing efficient attention studies mainly address the following question:

How can attention be approximated more efficiently while preserving segmentation accuracy?

In contrast, this work investigates a complementary yet largely unexplored problem:

Under what contextual conditions is attention approximation fundamentally appropriate for medical image segmentation?

Instead of introducing another efficient attention operator, we reformulate the dual-attention module of DA-TransUNet [7] into a controllable approximation framework and systematically investigate the interaction between approximation strategies, routing behavior, and task-specific contextual requirements. This perspective shifts the focus from designing increasingly efficient attention operators toward understanding the applicability and limitations of attention approximation itself.

III. METHOD

This section introduces the overall architecture of ApproxDA, then describes the three controllable approximation operators used to reformulate the dual-attention module, and finally analyzes their computational complexity.

A. Architecture Overview

ApproxDA-TransUNet adopts the hybrid CNN-ViT encoder-decoder structure of DA-TransUNet [7], with an R50-ViT-B/16 backbone pretrained on ImageNet-21K. Three CNN encoder stages produce multi-scale features at strides 4, 8, and 16; a 12-layer ViT processes 14×14 patch tokens. ApproxDABlocks replace all dual-attention modules at three skip connections—512 channels at 28×28 , 256 at 56×56 , and 64 at 112×112 —and at the bottleneck.

ApproxDA-TransUNet is designed around an explicit hypothesis: that three orthogonal approximation operators—windowed PAM, low-rank projection, and grouped CAM—can replace the quadratic DA module to yield both a practical efficient architecture and a systematic framework for studying when approximation is appropriate. Each operator targets a distinct approximation axis (spatial scope, representation fidelity, channel scope), so any two can be held fixed while the third varies. This makes ApproxDA not merely a faster model, but a model whose design enables controlled ablation of approximation effects across tasks with different contextual requirements.

Engineering fix. The publicly released DA-TransUNet code silently bypasses the 64-channel (112×112) skip connection via a guard condition that checks the wrong tensor dimension. If executed, `DANetHead(64, 64)` creates a `Conv2d` layer with a zero-element weight that causes `DataParallel` to raise an invalid-gradient error at the first backward call, confirmed on T4 $\times$ 2. Full $N \times N$ PAM at 112×112 ($N=12,544$) would also require ≈ 7 GB for the attention matrix alone. ApproxDA-TransUNet resolves both issues simultaneously.

Figure 3 illustrates the overall ApproxDA architecture.

B. Attention Approximation Formulation

The original DA module performs dense spatial and channel attention over the full feature map, with quadratic complexity



Fig. 3. ApproxDA-TransUNet architecture. ApproxDABlocks (Low-Rank Windowed PAM + Grouped CAM + gate routing) replace dual-attention modules at all skip connections and the bottleneck. `gate_mode` controls the routing during ablation.

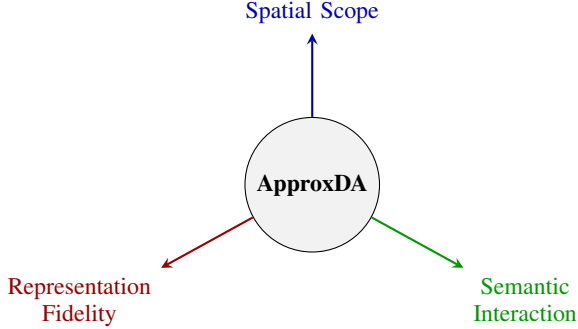


Fig. 4. ApproxDA's three-dimensional approximation space. Each axis is independently controllable. Spatial scope (M) most directly governs long-range context capture and thereby approximation safety on high-GCR tasks.

in both spatial resolution and channel dimension. ApproxDA reduces this cost along three independent axes, each targeting a distinct aspect of the approximation:

Spatial scope (window size M) controls *how far* PAM reaches. Full global attention ($M=H$) captures all long-range spatial relationships; restricting to $M \times M$ local windows trades long-range context for lower complexity. This axis most directly governs whether high-GCR tasks retain the anatomical context they require.

Representation fidelity (rank r) controls *how precisely* attention is computed within each window. Low-rank projection ($r \ll M^2$) approximates the affinity matrix at reduced cost; higher r approaches exact attention.

Semantic interaction (groups G) controls *which channels* may interact in CAM. Partitioning C channels into G independent groups reduces cross-channel dependency while retaining within-group semantic interaction.

These axes correspond to three distinct types of approximation: *spatial approximation* (how far attention reaches, controlled by M), *representation approximation* (how precisely affinity is computed, controlled by r), and *channel approximation* (how broadly channels interact, controlled by G). The axes are fully independent: any two of $\{M, r, G\}$ can be held fixed while the third varies, enabling clean ablations that isolate individual approximation effects. Figure 4 visualizes this three-dimensional approximation space.

C. Spatial Approximation

Instead of computing global position attention, ApproxDA partitions the feature map into non-overlapping windows of size $M \times M$. Attention is computed independently inside each local window, reducing complexity from $\mathcal{O}(N^2C)$ to $\mathcal{O}(NM^2C)$.

Window partitioning. Following [8], we partition the feature map into n_W non-overlapping $M \times M$ windows. Window clamping $M \leftarrow \min(M, H, W)$ allows $M=112$ to yield global attention at all scales, enabling a clean ablation that isolates windowing from low-rank as the performance bottleneck.

Low-rank projection. Following [9], query and key features inside each window are projected from $N_w=M^2$ to rank $r \ll M^2$ via a learned matrix $\mathbf{W}_r \in \mathbb{R}^{r \times N_w}$:

$$\tilde{\mathbf{C}} = \mathbf{W}_r \mathbf{C}^\top, \quad \tilde{\mathbf{D}} = \mathbf{W}_r \mathbf{D}^\top \in \mathbb{R}^{r \times C}. \quad (1)$$

The attention matrix $\mathbf{Q}\mathbf{K}^\top \in \mathbb{R}^{N_w \times N_w}$ is approximated by $\mathbf{Q}_r \mathbf{K}_r^\top \in \mathbb{R}^{N_w \times r}$, where $\mathbf{Q}_r, \mathbf{K}_r \in \mathbb{R}^{N_w \times r}$. Total complexity is $\mathcal{O}(NrC)$. Combined, the two approximations reduce attention storage at 112×112 from ≈ 15 GB to ≈ 38 MB ($\sim 390\times$).

These two hyperparameters—window size M and projection rank r —jointly constitute the spatial approximation operator and form the primary experimental axes investigated on Synapse.

D. Channel Approximation

The original Channel Attention Module computes a $C \times C$ channel attention matrix at $\mathcal{O}(C^2N)$ cost. ApproxDA approximates this by partitioning C channels into G groups of width $C_g = C/G$, computing independent per-group attention:

$$X_{ji}^{(g)} = \frac{\exp(\mathbf{A}_i^{(g)} \cdot \mathbf{A}_j^{(g)})}{\sum_{k=1}^{C_g} \exp(\mathbf{A}_k^{(g)} \cdot \mathbf{A}_j^{(g)})}, \quad (2)$$

with total cost $\mathcal{O}(C^2N/G)$. Compared with dense channel interaction, grouped attention significantly reduces computational complexity while preserving local channel dependency. Grouped decomposition also eliminates an integer-division underflow that occurs in the original DANetHead at $C=64$.

The grouping number G provides another controllable approximation dimension, allowing the influence of channel approximation to be studied independently from spatial approximation.

E. Routing Strategy

The learnable routing mechanism of DA-TransUNet [7] is retained and extended to a configurable `gate_mode`. The outputs of the spatial and channel attention branches are fused via:

$$\mathbf{y} = \mathbf{W}_f \left(g \cdot \hat{\mathbf{P}} + (1 - g) \cdot \hat{\mathbf{C}} \right) + \mathbf{x}, \quad (3)$$

where $\hat{\mathbf{P}}$ and $\hat{\mathbf{C}}$ are the normalized PAM and CAM outputs, $\mathbf{W}_f$ is a 1×1 convolution, and g is determined by the `gate_mode` parameter:

TABLE I
THEORETICAL COMPLEXITY OF ATTENTION MODULES

Module	Original	Proposed
PAM	$\mathcal{O}(N^2C)$	$\mathcal{O}(NrC)$ [$r \ll N$]
CAM	$\mathcal{O}(C^2N)$	$\mathcal{O}(C^2N/G)$
Routing	Fixed sum	Configurable gate

pam: $g = 1.0$ (PAM only)
 cam: $g = 0.0$ (CAM only)
 fixed: $g = 0.5$ (equal blend)
 learn: $g = \sigma(\mathbf{W}_g \cdot \text{GAP}(\mathbf{x})) \in (0, 1)^C$

This single-parameter design enables systematic ablation of routing strategy without any other architectural change. ApproxDA does not treat this routing module as a primary architectural contribution; instead, it serves as an analyzable component whose optimization behavior is investigated in Section V. As demonstrated later, the routing coefficient consistently converges to an almost constant value, revealing an unexpected optimization phenomenon.

F. Loss Function

The training objective combines Dice loss and cross-entropy loss with equal weights:

$$\mathcal{L} = \frac{1}{2}\mathcal{L}_{\text{Dice}} + \frac{1}{2}\mathcal{L}_{\text{CE}}. \quad (4)$$

G. Complexity Analysis

Table I summarizes the theoretical complexity reduction of the proposed approximation. Windowed PAM reduces spatial attention complexity from $\mathcal{O}(N^2C)$ to $\mathcal{O}(NM^2C)$, and low-rank projection further reduces it to $\mathcal{O}(NrC)$ where $r \ll C$. Grouped CAM reduces channel interaction from $\mathcal{O}(C^2N)$ to $\mathcal{O}(C^2N/G)$.

It should be emphasized that these reductions correspond to layer-level attention complexity rather than end-to-end network complexity. Since convolutional and transformer backbone operations still dominate the overall computation, theoretical complexity reduction does not necessarily translate into proportional reductions in whole-network FLOPs. Instead, ApproxDA provides a controllable approximation formulation that enables systematic investigation of the trade-off between approximation strength and segmentation performance.

IV. EXPERIMENTS

We evaluate ApproxDA-TransUNet on three medical image segmentation benchmarks spanning different global context requirements. The experiments systematically examine computational efficiency, task-dependent approximation behavior across GCR regimes, the dynamics of symmetric gate routing, and the relationship between approximation strength and task context.

TABLE II
DATASET PROPERTIES AND GLOBAL CONTEXT REQUIREMENT (GCR)

Dataset	Classes	Object Type	Long-range Context	GCR
Synapse	8	Multi-organ CT	High (inter-organ)	High
Kvasir-SEG	1	Polyp	Low (local boundary)	Low
ISIC 2018	1	Skin lesion	Low (texture/edge)	Low

High-GCR tasks require cross-organ anatomical reasoning; low-GCR tasks are dominated by local appearance cues. GCR governs approximation safety (Section V).

A. Experimental Setup

Datasets. We evaluate ApproxDA on three representative medical image segmentation benchmarks covering different contextual characteristics.

Synapse Multi-Organ CT [17]: 30 abdominal CT scans with an 18/12 train/test split covering 8 organ classes (9 classes with background). Evaluation metrics: mean DSC and HD95.

Kvasir-SEG [18]: 1000 endoscopy polyp images (binary segmentation) with an 800/200 train/test split. Evaluation metrics: DSC and mIoU.

ISIC 2018 [19]: 2594 dermoscopy skin lesion images (binary segmentation) with a 2075/519 train/test split (80/20). Evaluation metrics: DSC and mIoU.

Table II summarizes the three datasets alongside their contextual characteristics and GCR classification, which motivates the cross-task analysis in Section V.

Implementation Details. All experiments use an R50-ViT-B/16 backbone pretrained on ImageNet-21K. Training uses SGD (momentum 0.9, weight decay 10^{-4}), initial learning rate 0.01 with polynomial decay, batch size 24, image size 224×224 , 300 epochs, and seed 1234. The DA-TransUNet baseline runs on T4 $\times$ 1. ApproxDA-TransUNet runs on T4 $\times$ 2 via `torchrun` DDP with per-GPU batch 12 and NCCL all-reduce, under identical total batch size and base learning rate. Default ApproxDA configuration: $M=7$, $r=32$, $G=8$. Our DA-TransUNet re-run achieves validation DSC 0.795, consistent with the paper-reported 0.798, confirming faithful replication.

B. Computational Efficiency of ApproxDA

Table III compares the computational characteristics of ApproxDA and DA-TransUNet on Synapse. Although per-GPU VRAM is reduced substantially—from 11.5 GB to 6.4 GB—total GFLOPs are slightly higher (32.1 vs. 30.2) because the shared ViT backbone dominates total computation and Conv1d/Linear projection overhead partially offsets decoder-level savings. The reduced memory footprint enables multi-GPU distributed training via `torchrun` DDP, resolving the DataParallel failure described in Section III.

C. Task-dependent Behavior of Attention Approximation

We examine whether efficient attention approximation behaves consistently across medical segmentation tasks with different contextual requirements. Tables IV–VI compare

TABLE III
EFFICIENCY COMPARISON ON SYNAPSE (T4). GFLOPS MEASURED BY `FV_CORE` (SINGLE 224×224 SLICE, INCLUDES ATTENTION BMM).

Method	Params (M)	GFLOPs	Train VRAM	Train Time	Infer (s/vol)	Multi-GPU
DA-TransUNet [7]	107.95	30.2	11.5 GB	11.4h $\times$ 1	120.0	No [†]
ApproxDA (gate=pam, $M=7$, $r=32$)	112.98	32.1	6.4 GB/GPU	8.3h $\times$ 2	121.3	Yes (DDP)
ApproxDA (gate=learn, $M=7$, $r=32$)	114.90	32.1	10.6 GB	11.5h $\times$ 1	114.2	Yes (DDP)

[†]DA-TransUNet crashes DataParallel on T4 $\times$ 2 due to a zero-element weight in DANetHead(64,64). ApproxDA uses DDP (torchrun, NCCL) which avoids this failure. DDP: per-GPU batch 12 (total 24), LR unchanged.

TABLE IV
SEGMENTATION RESULTS ON SYNAPSE MULTI-ORGAN CT

Method	DSC (%) $\uparrow$	HD95 (mm) $\downarrow$
<i>CNN-based</i>		
U-Net [1]	76.85	39.70
U-Net++ [2]	76.91	36.93
Residual U-Net [4]	76.95	38.44
Att-Unet [3]	77.77	36.02
MultiResUNet [5]	77.42	36.84
<i>Transformer-based</i>		
TransUNet [12]	77.48	31.69
UCTransNet [16]	78.23	26.75
TransNorm [20]	78.40	30.25
MIM [21]	78.59	26.59
Swin-Unet [13]	79.13	21.55
DA-TransUNet [7]	79.80	23.48
<i>Dual-attention approximation (ours)</i>		
ApproxDA-TransUNet	78.64	31.09

Baselines from [7] Table 1 (SGD, 500ep, 224×224). ApproxDA: gate=pam, $M=7$, $r=32$, $G=8$, SGD 300ep. We also re-ran DA-TransUNet under our protocol (SGD, 300ep) and obtained 72.03% DSC; the gap from paper-reported (79.80%) is primarily due to optimizer and epoch differences (Adam/500ep vs. SGD/300ep), so we report paper-reported values for all Synapse baselines.

ApproxDA-TransUNet against DA-TransUNet and established baselines on all three benchmarks.

Synapse. Table IV compares DSC and HD95 against 11 published methods. CNN-based methods achieve 76–78% DSC. Transformer-based methods progressively improve global context modeling, with Swin-Unet (79.13%) and DA-TransUNet (79.80%) topping the leaderboard. ApproxDA-TransUNet (gate=pam, $M=7$, $r=32$) achieves 78.64% DSC and 31.09 mm HD95, surpassing all CNN-based baselines; the -1.16% gap to DA-TransUNet indicates that high-GCR tasks require richer global spatial interaction than the approximated model can provide.

Kvasir-SEG. Table V compares polyp segmentation results against all six methods reported in [7]. We re-ran DA-TransUNet under the same SGD/300ep protocol as ApproxDA to enable a direct comparison, obtaining 88.44% DSC, 81.70% mIoU, and 53.04 mm HD95. ApproxDA-TransUNet (gate=learn) achieves 89.24% DSC, 83.40% mIoU, and 42.60 mm HD95—outperforming the reproduced DA-TransUNet baseline (+0.80% DSC, +1.70% mIoU, -10.44 mm HD95).

ISIC 2018. Table VI reports skin lesion segmentation results

TABLE V
SEGMENTATION RESULTS ON KVASIR-SEG POLYP

Method	DSC (%) $\uparrow$	mIoU (%) $\uparrow$	HD95 (mm) $\downarrow$
U-Net [1]	88.22	80.12	—
Att-Unet [3]	86.61	78.01	—
U-Net++ [2]	86.57	77.67	—
Res-UNet [4]	77.85	66.04	—
TransUNet [12]	87.91	80.03	—
DA-TransUNet [7] [†]	88.44	81.70	53.04
ApproxDA-TransUNet (ours)	89.24	83.40	42.60

Other baselines from [7] Table 2 (Adam, 500ep, 75/25 split); HD95 not reported. [†]DA-TransUNet re-run under the same protocol as ApproxDA (SGD, 300ep, 80/20 split). Paper-reported [7]: 88.47% DSC, 81.02% mIoU.

TABLE VI
SEGMENTATION RESULTS ON ISIC 2018 SKIN LESION

Method	DSC (%) $\uparrow$	mIoU (%) $\uparrow$
U-Net [1]	87.22	81.14
Att-Unet [3]	87.60	81.51
U-Net++ [2]	87.30	81.33
Res-UNet [4]	83.32	76.51
TransUNet [12]	88.78	82.63
DA-TransUNet [7]	88.88	82.78
ApproxDA-TransUNet (ours)	<i>pend.</i>	<i>pend.</i>

Baselines from [7] Table 2 (Adam, 50ep, 75/25 split). ApproxDA: SGD, 300ep, 80/20 split.

against the same six baselines from [7]. Our ApproxDA experimental results are pending completion.

Strikingly, ApproxDA yields opposing effects across datasets: it consistently underperforms on Synapse (-1.16% DSC) while achieving competitive or improved performance on Kvasir (+0.80% DSC). This contrast is not attributable to hyperparameter differences—both experiments use the same M , r , and G —and instead reflects the intrinsically different degree to which each task relies on long-range spatial context.

D. Analysis of Symmetric Gate Collapse

Although the learnable gate (gate=learn) was designed to dynamically balance PAM and CAM branches, Table VII reveals that it consistently ranks last on Synapse. The gate coefficient converges to $g \approx 0.5$ and remains nearly constant throughout training, producing a fixed 50/50 blend that performs below both PAM-only (78.64%) and CAM-only (78.26%)

TABLE VII
GATE MODE ABLATION ON SYNAPSE ($M=7$, $r=32$ UNLESS NOTED)

Gate Mode	g	DSC (%) $\uparrow$	HD95 (mm) $\downarrow$
gate=pam	1.0	78.64	31.09
gate=cam	0.0	<u>78.26</u>	<u>30.59</u>
gate=learn ($M=14$, $r=64$)	$\approx 0.5^*$	78.04	29.09
gate=learn	$\approx 0.5^*$	77.78	34.29

All rows: ApproxDA-TransUNet. *Gate collapsed to $g \approx 0.5$ throughout training; see Section V-A. **Bold**: best; underline: second best per column.

TABLE VIII
SPATIAL APPROXIMATION ABLATION ON SYNAPSE ($G=8$)

Config	M	r	Gate	DSC (%) $\uparrow$	HD95 (mm) $\downarrow$
Windowed + low-rank	7	32	pam	78.64	31.09
Windowed + low-rank	14	64	learn [†]	78.04	29.09
Global + low-rank	112	64	pam	78.93	31.21

[†]Gate collapsed to $g \approx 0.5$; gate mode differs from the other rows. $M=112$ via window clamping yields full-map attention at all scales. The +0.29% DSC gain ($M=7 \rightarrow 112$, gate=pam) indicates low-rank projection is the binding constraint: even global attention ($r=64$) leaves a -0.87% gap to DA-TransUNet. Note: r co-varies with M ; rows are not single-variable ablations.

configurations. This collapse is observed across both $M=7$ and $M=14$, confirming that it is a property of symmetric two-branch optimization rather than a consequence of limited receptive field. The gradient-level derivation is provided in Section V-A.

E. Understanding Approximation Safety

Which approximation axis drives the performance gap?

Table VIII isolates the contribution of spatial windowing by comparing windowed ($M=7$) against effectively global ($M=112$, window-clamped) attention under gate=pam. The $M=14$ result from the gate ablation (gate=learn) is also shown for reference to provide a denser sampling of window sizes.

Removing windowing yields only +0.29% DSC improvement, while the -0.87% gap to DA-TransUNet persists under full-map attention. This reveals that low-rank projection—not spatial windowing—is the deeper bottleneck on high-GCR tasks: even global attention with $r=64$ cannot recover the attention fidelity lost by low-rank decomposition.

Cross-task summary. Table IX consolidates results across all benchmarks. The direction of approximation’s effect reverses with GCR: -1.16% DSC on high-GCR Synapse versus $+0.80\%$ DSC on low-GCR Kvasir-SEG. Multi-organ CT requires cross-organ anatomical reasoning that spatial approximation disrupts, while polyp segmentation depends primarily on local boundary contrast where windowed attention acts as an implicit regularizer. Figure 5 illustrates this context-dependent pattern qualitatively.

Together, the controlled experiments across datasets, gate modes, and window sizes establish a consistent picture: attention approximation behaves differently depending on the

TABLE IX
CROSS-TASK APPROXIMATION SAFETY VS. GLOBAL CONTEXT REQUIREMENT

Dataset	GCR	DA-TransUNet [†]	Best ApproxDA	Δ DSC
Synapse	High	79.80%	78.64% ^a	-1.16%
Kvasir-SEG	Low	88.44% ^c	89.24% ^b	$+0.80\%$
ISIC 2018	Low	88.88%	<i>pend.</i>	<i>pend.</i>

[†]Paper-reported [7] except Kvasir-SEG which is our re-run (SGD, 300ep, 80/20 split). ^agate=pam. ^bgate=learn. ^cOur re-run; paper-reported: 88.47%.



Fig. 5. Qualitative segmentation comparison. **Top (Synapse)**: DA-TransUNet preserves multi-organ boundaries while ApproxDA shows degradation on inter-organ regions, illustrating high-GCR sensitivity. **Bottom (Kvasir-SEG)**: ApproxDA matches or exceeds DA-TransUNet on polyp boundaries, illustrating low-GCR tolerance. [Generated after experiments complete.]

global context requirements of the target task. These observations motivate the formal GCR analysis in Section V.

V. DISCUSSION

A. Observation 1: Gate Collapse

Derivation. Given fused output $\mathbf{F} = g\mathbf{F}_P + (1 - g)\mathbf{F}_C$, the gate gradient is:

$$\frac{\partial \mathcal{L}}{\partial g} = \frac{\partial \mathcal{L}}{\partial \mathbf{F}} \cdot (\mathbf{F}_P - \mathbf{F}_C). \quad (5)$$

At initialization $g \approx 0.5$, both branches receive equal gradient $\frac{1}{2} \partial \mathcal{L} / \partial \mathbf{F}$. Without incentive to differentiate, $\mathbf{F}_P \approx \mathbf{F}_C$, so Eq. (5) $\approx \mathbf{0}$. This is self-reinforcing: equal gradients $\rightarrow$ convergent representations $\rightarrow$ near-zero gate gradient $\rightarrow g$ fixed at 0.5. This represents a **stable equilibrium of symmetric dual-branch optimization**: once reached, no first-order gradient signal can escape it.

Empirical confirmation. After 300 epochs, the gate range at $M=7$ is $[0.4993, 0.5010]$ with $\Delta g = 0.0017$. At $M=14$: $g \approx 0.5002$ with $\Delta g \approx 0.0000$. Collapse is identical at both window sizes, confirming it is a fundamental property of symmetric two-branch gating and not a consequence of limited receptive field. Figure 6 visualizes the collapse trajectory.

Design implication. Symmetric two-branch gating is an unstable fixed point that any dual-attention architecture will converge to without explicit symmetry breaking. Future designs should initialize branches asymmetrically, use diversity-encouraging loss terms, or replace the gate with a non-

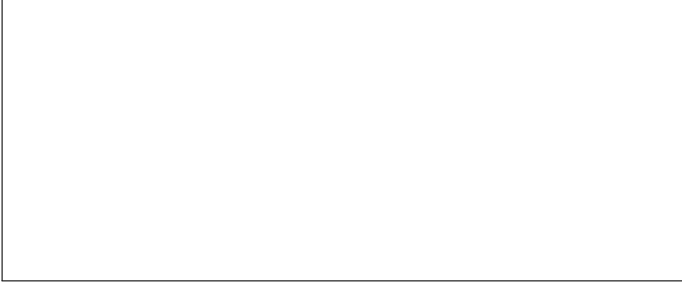


Fig. 6. Gate value g vs. training epoch for $M=7$ and $M=14$. Both converge to $g \approx 0.5$ by epoch 100, independent of window size. Shaded band shows gate range across network blocks. [Generated from training logs after experiments complete.]

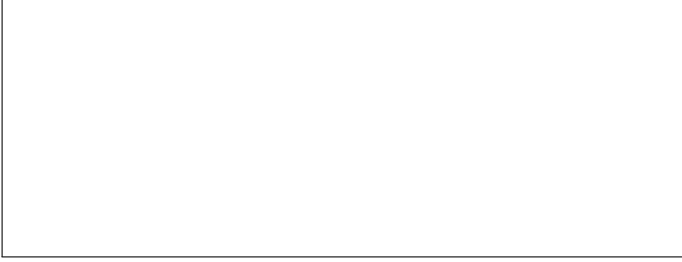


Fig. 7. Gate value vs. per-block input entropy scatter (300-epoch checkpoint, Synapse validation). Gate values cluster tightly at $g \approx 0.5$ regardless of input entropy, confirming no sample-adaptive routing occurs. [Generated from inference analysis.]

symmetric routing mechanism to escape this degenerate equilibrium.

B. Observation 2: Task-Dependent Approximation Safety

Although ApproxDA adopts the same approximation operators across all experiments, its effectiveness varies considerably among different segmentation datasets. ApproxDA consistently underperforms the original dual-attention module on Synapse, while achieving competitive or improved performance on Kvasir.

We attribute this phenomenon to the different global contextual requirements of the underlying segmentation tasks. Multi-organ abdominal CT segmentation requires reasoning over long-range anatomical relationships among multiple organs. Aggressive attention approximation inevitably weakens such global dependency modeling, leading to performance degradation.

In contrast, polyp and skin lesion segmentation are primarily determined by local appearance, boundary contrast, and texture consistency. Consequently, restricting attention to local regions introduces only limited information loss while simultaneously acting as an implicit regularizer, which explains the improved generalization observed on Kvasir. This interpretation motivates a more precise characterization in terms of Global Context Requirement, formalized below.

C. GCR and Approximation Safety

We formally define two complementary task-level properties:

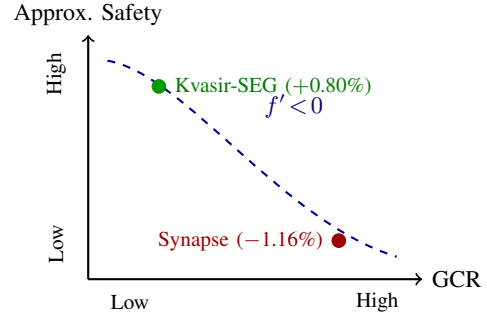


Fig. 8. Conceptual relationship between GCR and approximation safety (Eq. (6)). The dashed curve is hypothetical, not fitted; confirmed data points for Kvasir-SEG (low GCR) and Synapse (high GCR) are consistent with the monotonically decreasing trend. ISIC 2018 result pending.

Global Context Requirement (GCR) is the degree to which accurate segmentation depends on long-range spatial dependencies between distant anatomical elements. GCR is a latent task property, intrinsic to the task and independent of model architecture. High-GCR tasks (multi-organ CT) require cross-organ anatomical reasoning—liver position relative to spleen, aorta relative to vena cava—where disrupting global spatial interaction degrades delineation. Low-GCR tasks (polyp, skin lesion) are dominated by local boundary contrast and texture, with minimal dependence on distant context.

Approximation Safety is the degree to which spatial attention approximation can be introduced without degrading segmentation accuracy on a given task.

We hypothesize an inverse relationship between these properties, which we treat as a predictive claim rather than a proven causal law:

$$\text{Approximation Safety} \approx f(\text{GCR}), \quad f'(\cdot) < 0. \quad (6)$$

Equation (6) is a conceptual claim, not an analytic proof. It summarizes the empirical pattern: the same ApproxDA configuration achieves +0.80% DSC on low-GCR Kvasir-SEG and −1.16% on high-GCR Synapse—opposite effects consistent with a monotonically decreasing relationship. Figure 8 plots the confirmed data points against this hypothesis.

Attention approximation should not be regarded as a binary design choice but as a space whose safety *appears to depend* on GCR. Different approximation configurations (Figure 4) suit different GCR regimes. The gate=learn improvement on Kvasir (+0.80%) further suggests that on low-GCR tasks, the implicit regularization of a collapsed 50/50 gate may confer a generalization benefit that PAM-only routing does not. That the same collapsed routing strategy helps low-GCR tasks but hurts high-GCR tasks underscores that the fixed equilibrium is not inherently harmful—its effect is task-dependent.

D. Design Principles

The GCR framework and the gate collapse finding together provide three practical principles for efficient attention design in medical image segmentation.

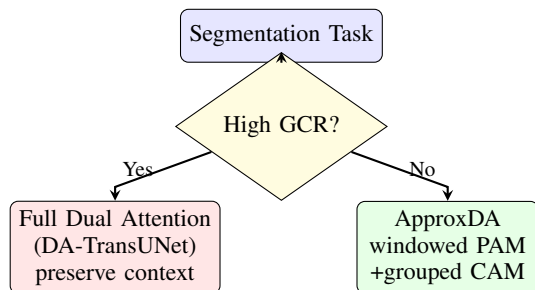


Fig. 9. Practical design guideline derived from GCR analysis. Tasks requiring cross-organ anatomical reasoning (high GCR) should retain full dual attention. Local-structure tasks (low GCR) can safely use ApproxDA, reducing per-GPU VRAM by 44% with no accuracy penalty.

First, attention approximation should be selected according to the contextual characteristics of the target segmentation task, rather than applied uniformly across all medical imaging applications.

Second, adaptive routing mechanisms require sufficient feature diversity between attention branches; otherwise, gradient symmetry may cause routing collapse regardless of window size or rank.

Finally, future efficient attention architectures should explicitly consider approximation safety together with computational efficiency, instead of focusing exclusively on reducing theoretical complexity. Figure 9 summarizes these guidelines as a practical decision tree.

VI. CONCLUSION

This paper presented ApproxDA, a controllable approximation formulation derived from the dual-attention module of DA-TransUNet [7]. Rather than proposing another efficient attention operator, ApproxDA was designed to systematically investigate the effectiveness of attention approximation under different medical image segmentation tasks.

Extensive experiments demonstrate that attention approximation is fundamentally context dependent. While approximation remains effective for local-structure-dominant segmentation tasks such as polyp segmentation (ApproxDA achieves 89.24% DSC on Kvasir-SEG, +0.80% over the full-attention baseline), it consistently degrades performance on tasks requiring extensive global anatomical reasoning such as multi-organ CT segmentation (−1.16% DSC on Synapse).

Furthermore, our analysis reveals that the conventional learnable routing mechanism naturally collapses because of gradient symmetry, causing adaptive routing to degenerate into an almost fixed fusion strategy. This collapse is independent of window size or rank, representing a fundamental limitation of symmetric two-branch gating rather than a windowing artifact.

Together, these findings suggest Global Context Requirement (GCR) as a useful predictive factor for whether attention approximation will be beneficial or harmful for a given segmentation task—shifting the research question from *how to approximate* to *when to approximate*.

Future efficient attention research should move beyond designing increasingly lightweight attention operators toward understanding when approximation is fundamentally appropriate. We hope this study provides practical guidelines for designing future efficient attention architectures in medical image segmentation. In future work, we plan to investigate routing mechanisms that explicitly preserve branch diversity, as well as broader approximation strategies for foundation medical vision models. This work takes a first step toward answering *when*—not merely *how*—to approximate attention in medical image segmentation.

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